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ADSORBING MECHANISM AND ADSORBING DEVICE WITH THE SAME

Abstract

The present disclosure provides an adsorbing mechanism and an adsorbing device with the same. The adsorbing mechanism includes: a base, a plurality of suction cup units disposed on the base, and a clamping unit. The suction cup unit is used to telescopically move along a first direction, and the suction cup unit is connected to an external vacuum device, such that negative pressure is generated in the suction cup units for adsorbing a product. The clamping unit disposed on a side of the base along the first direction. The clamping unit includes a first clamping member and a second clamping member. Wherein at least one of the first clamping member and the second clamping member is configured to move close to or away from each other to clamp or release the suction cup units.

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Background/Summary

FIELD

[0001] The subject matter relates to adsorption of products, and more particularly, to an adsorbing mechanism and an adsorbing device with the adsorbing mechanism.

BACKGROUND

[0002] In actual production and processing, products need to be transferred between different processing stations. During the transfer of the product, a clamping mechanism or an adsorbing mechanism is used to secure the product. The adsorbing mechanism, such as an electrostatic chuck, may be suitable to be used in a limited space.

[0003] However, when the product has an irregular surface, such as a convex or concave surface, the existing adsorbing mechanism may not be fully attached to the product, which may cause the product to fall and break during the adsorption and transferring process.

[0004] Therefore, there is room for improvement in the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Implementations of the present technology will now be described, by way of example only, with reference to the attached figures.

[0006] FIG. 1 is a diagrammatic view of an adsorbing device according to an embodiment of the present disclosure.

[0007] FIG. 2 is a diagrammatic view of an adsorbing mechanism of the adsorbing device in FIG. 1.

[0008] FIG. 3 is a diagrammatic view showing the adsorbing mechanism in FIG. 2, adsorbing a product.

[0009] FIG. 4 is a diagrammatic view showing the adsorbing mechanism in FIG. 3, lifting up the adsorbed product.

[0010] FIG. 5 is a diagrammatic view of a suction cup unit of the adsorbing mechanism in FIG. 2.

DETAILED DESCRIPTION

[0011] It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different FIG.s to indicate corresponding or analogous components. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures, and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale and the proportions of certain parts may be exaggerated to better illustrate details and features of the present disclosure.

[0012] The term “comprising,” when utilized, means “including, but not necessarily limited to”; it specifically indicates open-ended inclusion or membership in the so-described combination, group,

series, and the like.

[0013] Referring to FIG. 1, an adsorbing device **1000** is provided according to an embodiment of the present disclosure. The adsorbing device **1000** is used to adsorb a product **200**. The adsorbing device **1000** includes an adsorbing mechanism **100** and a transferring mechanism **400**. The transferring mechanism **400** is used to drive the adsorbing mechanism **100** to move. The adsorbing device **1000** further includes an external vacuum device **600** connected to the adsorbing mechanism **100**. It can be understood that the adsorbing device **1000** may not include the external vacuum device **600**. That is, the external vacuum device **600** may be independent from the adsorbing device **1000**.

[0014] Referring to FIGS. 2 to 4, the adsorbing mechanism **100** is provided according to an embodiment of the present disclosure. The adsorbing mechanism **100** is used to adsorb the product **200**. The product **200** has an adsorbing surface **201**. In an embodiment, the adsorbing surface **201** of the product **200** can be a flat surface, a convex surface, a concave surface, and a stepped surface. The adsorbing mechanism **100** includes a base **10**, a plurality of suction cup units **20**, and a clamping unit **30**. The suction cup units **20** are arranged on the base **10** and parallel to each other. The suction cup units **20** can be telescopically arranged on the base **10**, that is, each suction cup unit **20** can telescopically move along a first direction Z relative to the base **10**. The suction cup unit **20** is connected to the external vacuum device **600**, which generates negative pressure in the suction cup unit **20** for adsorbing the adsorbing surface **201** of the product **200**. The clamping unit **30** is disposed on a side of the base **10** along the first direction Z. The clamping unit **30** is configured to clamp all of the suction cup units **20**.

[0015] In an embodiment, the base **10** may be a plate-shaped structure. The suction cup unit **20** extends through the base **10** and parallel to each other. The base **10** provides support for the suction cup units **20**.

[0016] In an embodiment, the suction cup units **20** are arranged in a row along a second direction X on the base **10**. The second direction X is perpendicular to the first direction Z. The suction cup units **20** are spaced apart from each other.

[0017] In an embodiment, the clamping unit **30** includes at least one first clamping member **32** and at least one second clamping member **34**. The first clamping member **32** and the second clamping member **34** are located on opposite sides of the suction cup units **20** along a third direction Y. The third direction Y is perpendicular to each of the first direction Z and the second direction X. It can be understood that the suction cup units **20** are located between the first clamping member **32** and the second clamping member **34**. The first clamping member **32** and the second clamping member **34** are all extend along the second direction X. At least one of the first clamping member **32** and second clamping member **34** can move close to or away from each other to clamp or release the suction cup units **20**. For example, the first clamping member **32** can move close to or away from the second clamping member **34**, or the second clamping member **34** can move close to or away from the first clamping member **32**, or both of the first clamping member **32** and the second clamping member **34** can move close to or away from each other.

[0018] Referring to FIGS. 1, 3 and 4, the adsorbing mechanism **100** is used for adsorbing the product **200** with a concave surface. That is, the concave surface is the adsorbing surface **201**, and the adsorbing mechanism **100** is used for adsorbing the adsorbing surface **201**. The product **200** is placed on a carrying platform **300**, and the adsorbing surface **201** faces the adsorbing mechanism **100**. The suction cup units **20** divided into two groups, which includes a first group and a second group. In the first direction Z, a projection of the first group of suction cup units **20** coincides with the product **200**, while a projection of the second group of suction cup units **20** is located outside the product **200**. For example, the number of the suction cup units **20** is nine, but is not limited thereto. The first group includes five suction cup units **20**, and the second group includes four suction cup units **20**. The number of the suction cup units **20** can be varied according to the size of the product **200** according to an actual situation.

[0019] The base **10**, the suction cup units **20**, and the clamping unit **30** are driven to approach the product **200**. The first group of suction cup units **20** is located directly above the product **200** and adsorb the product **200**. Since the first group of suction cup units **20** are telescopically mounted on the base **10**, the first group of suction cup units **20** can be telescopically adjusted according to the shape of the concave surface **201**, thereby ensuring that the first group of suction cup units **20** can fully adsorb the adsorbing surface **201**. The second group of suction cup units **20** move to the side of the product **200**. The first clamping member **32** and second clamping member **34** of the clamping unit **30** cooperate to clamp the suction cup units **20**. The external vacuum device **600** can provide negative pressure to the first group of the suction cup units **20** located directly above the product **200**, causing the first group of the suction cup units **20** to adsorb the product **200** under the negative pressure. The second group of suction cup units **20** can clamp the two opposite sides of the product **200** along the second direction X, thereby facilitating the transfer of the product **200**.

[0020] In an embodiment, the product **200** can be a concave mirror.

[0021] It is understandable that the external vacuum device **600** can independently provide negative pressure to one, two, or several suction cup units **20**, thereby reducing the air sources and saving costs.

[0022] In an embodiment, the suction cup units **20** are arranged in a row, with a gap between adjacent two suction cup units **20**. The gap is relatively small, resulting in tight arrangement of multiple suction cup units **20** without interference between them. In this way, the first group of suction cup units **20** can be arranged closely together to adapt to the shape of the adsorbing surface **201**, such that the first group of suction cup units **20** can fully adsorb the product **200**. At the same time, the second group of suction cup units **20** can also form an effective clamping force on the product **200**, thereby further enhancing the adsorption and clamping force on the product **200**.

[0023] Referring to FIGS. 2 and 5, the suction cup unit **20** includes a telescopic member **22** and a suction cup **24**. The telescopic member **22** is telescopically disposed on the base **10**. The telescopic member **22** includes a first end **221** and a second end **223** opposite to the first end **221**. The first end **221** of the telescopic member **22** is connected to the external vacuum device **600** (see FIG. 1), and the second end **223** of the telescopic member **22** is connected to the suction cup **24**. The telescopic member **22** is a hollow structure, and the suction cup **24** is connected to the external vacuum device **600** through the telescopic member **22**.

[0024] Referring to FIGS. 2 and 5, the telescopic member **22** includes a first cylinder **222**, a second cylinder **224**, and an elastic component **226**. The first cylinder **222** and the second cylinder **224** are substantially hollow structures. A first diameter “d1” of the first cylinder **222** is smaller than a second diameter “d2” of the second cylinder **224**. The first cylinder **222** extends through the base **10**. One end of the first cylinder **222** away from the second cylinder **224** is the first end **221** and connected to the external vacuum device **600** (see FIG. 1). One end of the second cylinder **224** away from the first cylinder **222** is the second end **222** and connected to the suction cup **24**. The second cylinder **224** is slidably sleeved on the first cylinder **222**. The second cylinder **224** is in sealed connection with the first cylinder **222**. The elastic member **226** is disposed between the first cylinder **222** and the second cylinder **224**. The second cylinder **224** is used to move relative to the first cylinder **222** to compress the elastic member **226** to enable the telescopic member **22** to have telescopic function.

[0025] The first clamping member **32** and the second clamping member **34** are located on opposite sides of the second cylinder **224**. Thereby, the telescopic member **22** can achieve the function of elastic telescoping. When the suction cup **24** abuts against the product **200**, the suction cup **24** drives the second cylinder **224** to move upward relative to the first cylinder **222** and compresses the elastic member **226**, causing the telescopic member **22** to be shortened. When the first group of suction cup **24** is separated from the product **200**, the elastic member **226** rebounds and drives the second cylinder **224** and the suction cup **24** to move downward relative to the first cylinder **222** to return to the original shape.

[0026] In an embodiment, the telescopic member **22** further includes a first fixing structure **225** and a second fixing structure **227**. The first fixing structure **225** is disposed on an outer surface of the first cylinder **222**. The second fixing structure **227** is disposed on an inner surface of the second cylinder **224**. Two end of the elastic member **226** are respectively fixed on the first fixing structure **225** and the second fixing structure **227**, so that the elastic member **226** can be disposed between the first cylinder **222** and the second cylinder **224**.

[0027] In an embodiment, the elastic member **226** may be a spring.

[0028] In an embodiment, a sealing ring **228** is arranged between the second cylinder **224** and the first cylinder **222**, so that the second cylinder **224** and the first cylinder **222** are connected in a sealed manner.

[0029] In an embodiment, an adaptive structure (not shown) such as a protrusion (not shown) and groove (not shown) can also be provided between the first cylinder **222** and the second cylinder **224** to prevent separation of the first cylinder **222** from the second cylinder **224**.

[0030] In an embodiment, the suction cup **24** may be made of soft material, thus the suction cup **24** can more closely adhere to the product **200**. In an embodiment, a material of the suction cup **24** may be silicone.

[0031] Referring to FIG. 2, the adsorbing mechanism **100** further includes a plurality

[0032] of hoses **40**, which are connected to the suction cup units **20** one-by-one. The hose **40** connects the external vacuum device **600** to the first end **221** of the telescopic member **22**. When the base **10**, the suction cup units **20**, and the clamping unit **30** move approach or away from the product **200**, the hose **40** can provide a certain space for the base **10**, the suction cup units **20**, and the clamping unit **30** to move, thereby ensuring that the suction cup units **20** are in communication with the external vacuum device **600** and improving the stability of vacuum adsorption.

[0033] The first clamping member **32** is fixed on one side of the suction cup units **20** and is spaced apart from the suction cup units **20**. There is a small gap between the first clamping member **32** and the suction cup units **20**. That is, the first clamping member **32** does not interfere with the second cylinder **224** during the second cylinder **224** move along the first direction Z. The second clamping member **34** is movably disposed on a side of the suction cup units **20** away from the first clamping member **32**. The second clamping member **34** can move close to or away from the first clamping member **32**. When the first group of suction cup units **20** adsorbs the product **200**, the second clamping member **34** can move close to the first clamping member **32** to clamp the suction cup units **20**. Thus, by limiting the first clamping member **32** to be fixedly arranged and the second clamping member **34** to be movably arranged, the power source is reduced, and the cost is saved.

[0034] In an embodiment, the clamping unit **30** includes one first clamping member

[0035] **32** and two second clamping members **34**. The two second clamping members **34** are parallel to each other. The two clamping members **34** are arranged at intervals along the first direction Z. Along the third direction Y, a projection of the first clamping member **32** is between projections of the two second clamping members **34**. That is, the first clamping member **32** and the two second clamping members **34** are arranged at intervals along the first direction Z. Thus, when the second clamping members **34** cooperate with the first clamping members **32** to clamp the suction cup units **20**, the two second clamping members **34** can abut against the suction cup units **20** with a larger area. Thereby, the clamping stability of the suction cup units **20** is improved, and the suction cup units **20** will not deflect.

[0036] In another embodiment, the clamping unit **30** can also include two or more than two first clamping member **32**, and more than two second clamping member **34**. For example, the number of the first clamping member **32** is two, and the number of the second clamping member **34** is three. The two first clamping members **32** and the three second clamping members **34** are arranged in a staggered manner along the first direction X. For another example, the number of the first clamping member **32** is three, and the number of the second clamping member **34** is four. The three first clamping members **32** and the four second clamping members **34** are arranged in a staggered

manner along the first direction Z. The specific number of the first clamping members **32** and the second clamping members **34** can be set according to the actual situation.

[0037] The clamping unit **30** further includes a driving member **36**. In an embodiment, the driving member **36** may be a linear cylinder. The driving member **36** is connected to the second clamping member **34**. The driving member **36** is used to drive the second clamping member **34** to move close to or away from the first clamping member **32**.

[0038] Thereby, the driving member **36** is provided to achieve the function of driving the second clamping member **34** to move.

[0039] In another embodiment, when the first clamping member **32** is movably arranged and the second clamping member **34** is fixedly arranged, the driving member **36** can be connected to the first clamping member **32**.

[0040] In yet another embodiment, when both the first clamping member **32** and the second clamping member **34** are movably arranged, the driving member **36** can be connected to the first clamping member **32** and the second clamping member **34**, respectively. For example, the driving member **36** may be a double-headed linear cylinder. The driving member **36** can drive the first clamping member **32** and the second clamping member **34** to approach or move away from each other.

[0041] Referring to FIG. 1, the adsorbing device **1000** further includes a connection base **500**. The connection base **500** is connected to the base **10** and the clamping unit **30** of the adsorbing mechanism **100**, as well as to the moving mechanism **400**. The moving mechanism **400** drives the adsorbing mechanism **100** to move through the connection base **500**. In an embodiment, the base **10** can be connected to the connection base **500**, the first clamping member **32** of the clamping unit **30** can be connected to the connection base **500**, the driving member **36** of the clamping unit **30** can be connected to the connection base **500**.

[0042] In an embodiment, the moving mechanism **400** may be a robot arm or a multi-axis linear module.

[0043] When the adsorbing mechanism **100** and adsorbing device **1000** are used for transferring product **200** with an irregular adsorbing surface **201**, the base **10**, the suction cup units **20**, and the clamping unit **30** are driven to move to the product **200**. The first group of suction cup units **20** is located directly above the product **200** and can abut against the adsorbing surface **201**. Then, the first group of suction cup units **20** can telescopically move according to the adsorbing surface **201** of the product **200**, thereby ensuring that the first group of suction cup units **20** can adsorb the adsorbing surface **201**. When the second group of suction cup units **20** moves to the sides of the product **200**, the base **10**, the suction cup units **20**, and the clamping unit **30** stop moving. The second group of suction cup units **20** forms a clamping force on the sides of the product **200**. When the base **10**, the multiple suction cup units **20**, and the clamping unit **30** stop moving, the first clamping member **32** and the second clamping members **34** of the clamping unit **30** clamp the suction cup units **20**, allowing the suction cup units **20** to maintain adsorption and clamping forces on the product **200** for easy transfer.

[0044] With the above configuration, the adsorbing mechanism **100** can attach to and adsorb the product **200** through the cooperation of the multiple suction cup units **20** and the clamping unit **30**, and the suction cup units **20** can also form a clamping force on the product **200**. Thereby, the adsorbing mechanism **100** can stably adsorb the product **200**, thereby preventing damages to the product **200** due to the product **200** being separating from the adsorbing mechanism **100**, and the waste of the product **200** may be reduced. In addition, since the suction cup unit **20** can telescopically move, the adsorbing mechanism **100** can be applied to products **200** with different adsorbing surfaces and materials. The adsorbing mechanism **100** has a wide range of applicability and can meet the production needs of a small number of diverse products **200**. The adsorbing mechanism **100** has a simple structure and low cost.

[0045] Even though information and advantages of the present embodiments have been set forth in

the foregoing description, together with details of the structures and functions of the present embodiments, the disclosure is illustrative only. Changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the present exemplary embodiments, to the full extent indicated by the plain meaning of the terms in which the appended claims are expressed.

Claims

1. An adsorbing mechanism comprising: a base; a plurality of suction cup units disposed on the base and parallel to each other, each of the plurality of suction cup units configured to telescopically move along a first direction, and the plurality of suction cup units further configured for connecting to an external vacuum device, such that a negative pressure is generated in the plurality of suction cup units for adsorbing an adsorbing surface of a product; and a clamping unit disposed on a side of the base along the first direction, the clamping unit comprising a first clamping member and a second clamping member, wherein the first clamping member and the second clamping member are located on opposite sides of the plurality of suction cup units, respectively, along a third direction, the third direction is perpendicular to the first direction, wherein at least one of the first clamping member and the second clamping member is configured to move close to or away from each other to clamp or release the plurality of suction cup units.
2. The adsorbing mechanism according to claim 1, wherein each of the plurality of suction cup units comprises a telescopic member and a suction cup, the telescopic member is telescopically disposed on the base, the telescopic member comprises a first end and a second end opposites to the first end, the first end is configured for connecting to the external vacuum device, the second end is connected to the suction cup, the telescopic member is a hollow structure, and the suction cup is connected to the external vacuum device through the telescopic member.
3. The adsorbing mechanism according to claim 2, wherein the telescopic member further comprises a first cylinder, a second cylinder, and an elastic component, the first cylinder extends through the base, one end of the first cylinder away from the second cylinder is the first end and connected to the external vacuum device, one end of the second cylinder away from the first cylinder is the second end and connected to the suction cup, the second cylinder is slidably sleeved on the first cylinder, the elastic member is disposed between the first cylinder and the second cylinder, the second cylinder is configured to move relative to the first cylinder to compress the elastic member, and the first clamping member and the second clamping member are located on opposite sides of the second cylinder along the third direction.
4. The adsorbing mechanism according to claim 3, wherein the telescopic member further comprises a first fixing structure and a second fixing structure, the first fixing structure is disposed on an outer surface of the first cylinder, the second fixing structure is disposed on an inner surface of the second cylinder, and two end of the elastic member are fixed on the first fixing structure and the second fixing structure, respectively.
5. The adsorbing mechanism according to claim 3, wherein the telescopic member further comprises a sealing ring disposed between the second cylinder and the first cylinder.
6. The adsorbing mechanism according to claim 1, wherein the first clamping member is fixed on one side of the plurality of suction cup units and spaced apart from the plurality of suction cup units, the second clamping member is movably disposed on a side of the plurality of suction cup units away from the first clamping member, and the second clamping member is configured to move close to or away from the first clamping member.
7. The adsorbing mechanism according to claim 6, wherein the clamping unit further comprises a driving member, the driving member is connected to the second clamping member, and the driving member is configured to drive the second clamping member to move close to or away from the first clamping member.

- 8.** The adsorbing mechanism according to claim 1, wherein the plurality of suction cup units is arranged in a row along a second direction, and the second direction is perpendicular to each of the first direction and the third direction.
- 9.** The adsorbing mechanism according to claim 1, wherein the clamping unit comprises one first clamping member and two second clamping members, the two second clamping members are parallel to each other, the two clamping members are spaced from each other along the first direction, along the third direction, an projection of the first clamping member is between projections of the two second clamping members.
- 10.** An adsorbing device comprising: an adsorbing mechanism comprising: a base; a plurality of suction cup units disposed on the base and parallel to each other, each of the plurality of suction cup units configured to telescopically move along a first direction, and the plurality of suction cup units further configured for connecting connected to an external vacuum device, such that a negative pressure is generated in the plurality of suction cup units for adsorbing an adsorbing surface of a product; and a clamping unit disposed on a side of the base along the first direction, the clamping unit comprising a first clamping member and a second clamping member, wherein the first clamping member and the second clamping member are located on opposite sides of the plurality of suction cup units, respectively, along a third direction, the third direction is perpendicular to the first direction, wherein at least one of the first clamping member and the second clamping member is configured to move close to or away from each other to clamp or release the plurality of suction cup units; and a transferring mechanism configured to drive the adsorbing mechanism to move.
- 11.** The adsorbing device according to claim 10, wherein each of the plurality of suction cup units comprises a telescopic member and a suction cup, the telescopic member is telescopically disposed on the base, the telescopic member comprises a first end and a second end opposites to the first end, the first end is configured for connecting to the external vacuum device, the second end is connected to the suction cup, the telescopic member is a hollow structure, and the suction cup is connected to the external vacuum device through the telescopic member.
- 12.** The adsorbing device according to claim 11, wherein the telescopic member further comprises a first cylinder, a second cylinder, and an elastic component, the first cylinder is extends through the base, one end of the first cylinder away from the second cylinder is the first end and connected to the external vacuum device, one end of the second cylinder away from the first cylinder is the second end and connected to the suction cup, the second cylinder is slidably sleeved on the first cylinder, the elastic member is disposed between the first cylinder and the second cylinder, the second cylinder is configured to move relative to the first cylinder to compress the elastic member, and the first clamping member and the second clamping member are located on opposite sides of the second cylinder along the third direction.
- 13.** The adsorbing device according to claim 12, wherein the telescopic member further comprises a first fixing structure and a second fixing structure, the first fixing structure is disposed on an outer surface of the first cylinder, the second fixing structure is disposed on an inner surface of the second cylinder, and two end of the elastic member are fixed on the first fixing structure and the second fixing structure, respectively.
- 14.** The adsorbing device according to claim 12, wherein the telescopic member further comprises a sealing ring disposed between the second cylinder and the first cylinder.
- 15.** The adsorbing device according to claim 10, wherein the first clamping member is fixed on one side of the plurality of suction cup units and spaced apart from the plurality of suction cup units, the second clamping member is movably disposed on a side of the plurality of suction cup units away from the first clamping member, and the second clamping member is configured to move close to or away from the first clamping member.
- 16.** The adsorbing device according to claim 15, wherein the clamping unit further comprises a driving member, the driving member is connected to the second clamping member, and the driving

member is configured to drive the second clamping member to move close to or away from the first clamping member.

17. The adsorbing device according to claim 10, wherein the plurality of suction cup units are arranged in a row along a second direction, the second direction is perpendicular to each of the first direction and the second direction.

18. The adsorbing device according to claim 10, wherein the clamping unit comprises one first clamping member and two second clamping member, the two second clamping members are parallel to each other, the two clamping members are spaced from each other along the first direction, along the third direction, an projection of the first clamping member is between projections of the two second clamping members.

19. The adsorbing device according to claim 10, further comprising a connection base, wherein the connection base is connected to the base, the clamping unit, and the moving mechanism, the moving mechanism drives the adsorbing mechanism to move through the connection base.

20. The adsorbing device according to claim 10, wherein the moving mechanism is a robot arm or a multi-axis linear module.
